

Project Charter

Vision-Based Automated Attendance Management System Using Facial Recognition

1. Project Background

Attendance management is an essential administrative task in educational institutions. Traditional attendance methods such as manual roll calls, sign-in sheets, RFID cards, and QR code systems often require significant time and effort from instructors. These methods are also prone to errors, proxy attendance, and administrative overhead.

Recent advancements in Computer Vision and Artificial Intelligence have enabled automated facial recognition systems capable of identifying individuals in real-time. By leveraging these technologies, attendance can be recorded automatically with minimal human intervention.

This project aims to develop a Vision-Based Attendance Management System that utilizes facial recognition to automatically identify students and mark attendance using classroom images or live camera feeds.

2. Problem Statement

Educational institutions currently rely on manual or semi-automated attendance mechanisms that suffer from several limitations:

- Manual attendance consumes valuable class time.
- Human errors may occur during attendance recording.
- Proxy attendance remains a common issue.
- Attendance records require additional administrative effort.
- Existing attendance systems often require students to perform explicit actions such as scanning cards or QR codes.

3. Proposed Solution

The proposed system will use Computer Vision and Facial Recognition technologies to automate attendance marking. The system will maintain a database of registered students along with their facial embeddings and academic information such as Student ID, Name, Section, Department, and Semester.

During attendance sessions:

- The instructor opens the Flutter application.
- The classroom image or video feed is captured.
- The system detects faces present in the frame.
- Facial recognition algorithms identify registered students.
- Attendance is automatically recorded.
- Attendance reports are generated and stored.

4. Project Objectives

Primary Objectives

- Automate attendance marking using facial recognition.
- Reduce time spent on manual attendance processes.
- Improve attendance accuracy.
- Maintain digital attendance records.
- Provide attendance analytics and reporting.

Secondary Objectives

- Explore practical applications of Computer Vision.
- Develop a scalable cross-platform attendance application using Flutter.
- Ensure secure storage of student biometric information.

5. Target Audience

Primary Users

- University Students
- Faculty Members
- Course Instructors

Secondary Users

- Department Administrators
- Academic Coordinators
- Educational Institutions

6. Key Features

Feature 1: Student Registration Module

Students can register within the system by providing Student ID, Name, Department, Section, Semester, and Facial Images. The system generates facial embeddings for future recognition.

Feature 2: Face Recognition-Based Attendance

- Capture image or video feed.
- Detect faces within the frame.
- Match detected faces against registered students.
- Automatically mark attendance.

Feature 3: Attendance Dashboard

- View attendance records.
- Search attendance history.
- Filter by course or section.
- Track attendance percentages.

Feature 4: Attendance Reports

- Daily attendance reports.
- Weekly attendance summaries.
- Monthly attendance reports.
- Student-wise attendance statistics.

Feature 5: Course and Section Management

- Create courses.

- Assign sections.
- Manage enrolled students.
- View class-specific attendance.

Feature 6: Attendance Analytics

- Attendance trends.
- Attendance percentages.
- Frequently absent students.
- Class attendance summaries.

7. Scope of the Project

In Scope

- Student enrollment
- Facial image processing
- Face recognition
- Attendance marking
- Attendance reporting
- Dashboard visualization
- Database management

Out of Scope

- Student grading system
- Learning management system functionality
- Examination management
- Behavioral monitoring
- Emotion detection

8. Functional Requirements

Student Management

- Register students
- Update student information
- Delete student records

Attendance Management

- Capture attendance session
- Detect and recognize faces
- Mark attendance automatically
- Store attendance records

Reporting

- Generate reports
- Export attendance data
- View attendance statistics

9. Non-Functional Requirements

Performance

Recognition results should be generated within a few seconds.

Accuracy

The system should achieve high facial recognition accuracy under classroom conditions.

Security

Student biometric data must be securely stored. Access should be role-based.

Scalability

Support multiple classes and sections.

Usability

User-friendly interface for faculty members.

10. Potential Risks

Risk 1: Privacy Concerns

Facial data represents sensitive biometric information.

Mitigation: Store facial embeddings instead of raw images. Encrypt sensitive information. Implement access control.

Risk 2: Recognition Errors

Factors such as lighting, occlusion, and camera quality may affect recognition performance.

Mitigation: Use robust facial recognition models. Implement confidence thresholds.

Risk 3: Spoofing Attacks

Printed photographs or mobile images may attempt to bypass the system.

Mitigation: Implement basic liveness detection mechanisms.

Risk 4: Environmental Constraints

Classroom layout and seating positions may affect visibility.

Mitigation: Use high-resolution camera inputs. Capture multiple frames when required.

11. Expected Outcomes

- Automate attendance recording.
- Reduce attendance management effort.

- Improve operational efficiency.
- Provide digital attendance records and analytics.
- Demonstrate the practical application of Computer Vision and AI in educational environments.

12. Technology Stack

Frontend / Mobile

Flutter (Dart) — Cross-platform mobile and desktop application.

Backend

FastAPI / Django

Database

PostgreSQL

AI and Computer Vision

OpenCV, FaceNet, ArcFace, InsightFace

Deployment

Docker, AWS / Local Server

13. Success Criteria

- Attendance is marked automatically with high accuracy.
- Attendance reports are generated successfully.
- Faculty can manage attendance digitally.
- Recognition performance remains reliable under normal classroom conditions.
- The system demonstrates measurable reduction in attendance-taking time.
- Flutter application runs correctly on Android and iOS devices.